Expanding A3: the system architecture brief

A defines intended use and acceptance criteria; B configures the implementation; C assigns each service and decision to an accountable role.

Shared uncertainty-bearing record at every interface

Entity + variable + unit + spatial support + timestamp + model/version + evidence locator; distribution / interval / scenario + dependency links + validity status + upstream record IDs. Proposed schema.

MATERIAL TRANSFORMATIONS — retain the record, update the represented quantity

S1 GeoAI attributes → **S2** Material stock → **S3** Dated outflows → **S4** Usable recovery → **S5** Impact / decision

1 Data and identity service

Checks units, entity matches, missing values, age and lineage. Output: qualified inputs and flags; invalid records are held for review. Semantic linkage: Kosse et al. (2025).

2 Inference and calibration service

Observations + prior/model → state or parameter estimates with uncertainty. Preserve fitting/evaluation separation. Probabilistic formulation: Kapteyn et al. (2021).

3 Interface propagation service

For each $S_i \rightarrow S_{i+1}$: apply its transformation to joint samples, intervals or explicit scenarios. Output: downstream uncertainty plus dependency/lineage links. Check units, support, assumptions and lost fields. Proposed integration.

4 Diagnostic and validation service

Compare observations and predictions; evaluate residuals and, where applicable, interval coverage. Flag mismatch; investigate data, parameters or model form before revision. A flag does not identify its cause. Kapteyn et al. (2021).

5 Dependency and revision ledger

Version every estimate and link its upstream evidence. A revised input invalidates affected descendants; recalculate only after checks. Output: auditable S1–S5 change history. Proposed tracking service.

6 Decision and evidence interface

Expose impact ranges/scenarios, validity flags and sensitivity to recovery assumptions. Compare evidence with A; refer unresolved decisions to the assigned role. Role allocation: Agrawal et al. (2023).

Orchestration, timing and authority — not a substitute for uncertainty computation

Human or software actors schedule checks, request missing evidence, invoke services and record approvals. Multi-agent coordination distributes these tasks; agent-based simulation is a model choice. Agentic AI is an unverified optional proposal here. Updates may be periodic or event-triggered, according to intended use.

Responsibility test: every interface must name its input, transformation, uncertainty output and checking service.

Solid links show proposed information dependencies, not an empirically validated deployed architecture. Concept sources support individual functions; the complete S1–S5 integration is the review's synthesis.